Unit 8: Digitisation Planning and Quality Control

Digitisation converts information from physical or analogue sources into digital representations. A high-quality scan requires more than pressing the scanner button.

This unit connects planning, equipment, safe preparation, capture, quality control, OCR, documentation and secure delivery.

BLISS: digitising library collections for discovery. BRIM: digitising records while retaining completeness, context and access restrictions.

KEY POINT · Plan the service and safeguard the source before choosing capture settings.

Learning outcomes and practical evidence

Describe a repeatable digitisation workflow, and distinguish capture, quality assurance, quality control and verification.

Select appropriate equipment and justify capture settings for a particular document, photograph or bound volume.

Produce a scan manifest, check completeness and quality, verify simulated OCR, and document a safe, traceable release decision.

KEY POINT · The final output is a documented, checked and fit-for-purpose digital surrogate.

Why institutions digitise

Access: a searchable copy can serve students or authorised records users without repeated handling of a fragile original.

Preservation support: reducing handling may help protect originals, but digital files themselves require storage and preservation management.

Continuity and service: digital surrogates support remote discovery, internal administration and disaster preparedness where properly controlled.

KEY POINT · Digitisation helps access and care; it does not make conservation or recordkeeping obligations disappear.

Know the difference: digitisation versus digital preservation

Digitisation is creation of a digital representation of an analogue object or document.

Digital preservation is the ongoing work needed to keep digital information accessible, understandable and trustworthy over time.

For a scanned certificate, digitisation produces images and text; preservation involves integrity checks, metadata, planned storage and future access.

KEY POINT · A scan is the beginning of a lifecycle, not the end.

A responsible eight-stage workflow

1. Select and assess materials; 2. confirm rights, access and scope; 3. inventory and prepare; 4. calibrate and capture.

5. Check image quality and sequence; 6. generate and review OCR where suitable; 7. describe, package and secure files; 8. preserve and provide authorised access.

A failed quality check sends work back to the appropriate earlier stage rather than being hidden in the final folder.

KEY POINT · Every stage should produce evidence or a documented decision.

Selection and prioritisation

Select materials based on user need, institutional mandate, significance, condition, rights and expected benefit.

Estimate volume, page or item types, staffing, storage, equipment, quality requirements and processing time before promising delivery.

Do not make highly restricted case files or delicate bindings a casual first project; seek authorisation and specialist handling.

KEY POINT · Prioritisation is a risk-and-value decision, not simply a count of pages.

Define a digitisation specification

A specification states the source category, capture equipment, resolution method, colour mode, file format, image quality and naming rules.

Also define sequence, metadata, permitted edits, QA/QC method, retention of originals, security and who approves publication.

A local project specification must be chosen for the material and use; do not apply one resolution number to every object.

KEY POINT · Write down acceptance criteria before work starts.

Inventory and page-level control

Create a unique source ID and record title or description, box or shelf location, condition, rights and expected page/item count.

For multi-page sources, note covers, blank pages, foldouts, inserts and intended order; blank pages may be evidentially important.

Use identifiers in capture manifests rather than depending only on a technician remembering the sequence.

KEY POINT · A complete image set is one whose parts can be traced to an inventory.

Pre-capture handling and preparation

Inspect each item for tears, bindings, folds, loose fragments, staples and conditions requiring conservator advice.

Use clean workspaces, careful support and collection-approved handling procedures. Never force a fragile book flat for scanning.

Record any exceptional intervention or inability to capture a portion of the original rather than silently omitting it.

KEY POINT · The original is more valuable than completing a scan quickly.

Choosing capture equipment

Flatbed scanners are suitable for many loose, stable sheets; overhead cameras and book cradles can better support bound or fragile items.

Sheet-fed scanners can be efficient for approved robust loose papers, but may damage fragile or irregular materials.

Mobile-phone capture can support classroom demonstrations, but professional workflows must test geometry, focus, lighting and colour stability.

KEY POINT · Choose equipment by source condition and target quality, not speed alone.

Camera setup: light, geometry and focus

Keep the imaging plane parallel to the source where feasible; control skew, perspective, focus and glare.

Use stable, even illumination. Avoid reflections from glossy surfaces, harsh shadows and colour casts that obscure evidence.

Include scale and colour/tonal reference targets when required by the project specification, and document their role.

KEY POINT · A larger image cannot repair a blurred or poorly illuminated original capture.

Image resolution: PPI, pixel dimensions and source scale

For flat originals, pixels per inch (PPI) relates sample density to real physical size; pixel count = inches × selected PPI.

Example: a 6 × 4-inch photograph at 300 PPI yields approximately 1800 × 1200 pixels, or 2.16 million pixels.

More PPI does not guarantee better imaging; optics, focus, sensor quality, lighting and source detail also matter.

KEY POINT · Resolution is one measured property of image quality, not quality itself.

Worked calculation: uncompressed capture size

For an 8 × 10-inch image at 300 PPI: width = 2400 px; height = 3000 px; total = 7.2 million pixels.

At 24-bit RGB (3 bytes/pixel), uncompressed pixel payload = 7,200,000 × 3 = 21,600,000 bytes ≈ 21.6 MB (decimal).

File headers, colour profiles, compression and metadata change actual saved file size; this calculation is a baseline only.

KEY POINT · Compute the pixel payload separately from the size of a saved TIFF, PNG or JPEG file.

Colour mode and tonal information

Use colour capture when ink, highlights, stamps, paper colour or annotations carry meaning; greyscale may suit some monochrome sources.

Bit depth describes how many tonal values can be represented; choose a tested workflow appropriate to the source and institutional specification.

Do not convert a coloured seal or marginal note into black-and-white if doing so hides evidential or interpretive detail.

KEY POINT · Represent source information faithfully before thinking about smaller delivery copies.

Master and access derivatives

A capture master should retain the source information with minimal, controlled processing, with technical metadata and a clear chain of custody.

Access derivatives may be cropped, resized, compressed, redacted or watermarked for an approved purpose, without overwriting the master.

For example, retain a well-documented high-quality master while delivering a smaller JPEG or searchable PDF after review.

KEY POINT · Separate the preservation-quality source from convenience copies.

File naming and folder structure

Use stable item IDs, page sequence and role suffixes: LIB023_p001_master.tif, LIB023_p001_access.jpg.

Avoid names such as final2_NEW_reallyfinal.jpg: they prevent reliable matching, sorting and automation.

Keep the source manifest, processing log, approved derivatives and storage locations separate but clearly linked.

KEY POINT · A filename helps locate an item; metadata explains what it is.

What is quality assurance?

Quality assurance (QA) is the planned system for producing consistent quality: documented procedures, training, calibration and acceptance rules.

A QA plan states responsibilities, reference targets, sampling, acceptance thresholds, issue categories and escalation paths.

Quality control (QC) is the set of inspections and measurements performed on actual batches or items.

KEY POINT · QA designs the process; QC checks the outputs.

Quality-control checks: completeness first

Compare every output to the source inventory: count pages, verify sequence, check fronts/backs and note blank pages and inserts.

A perfectly sharp image of page 3 is not an acceptable replacement for a missing page 2.

Use explicit status values such as pass, rework, hold or not captured, with a reason and reviewer.

KEY POINT · A missing or duplicated page is a quality failure even when all other images are beautiful.

Quality-control checks: visual accuracy

Inspect sharpness, cropping, cut-off text, skew, glare, shadows, colour balance, exposure and readable small features.

Compare the image with the original or approved reference whenever possible. A cropped signature or marginalia can change meaning.

Automated image metrics can support checks, but they do not replace a reviewer who understands the content.

KEY POINT · Test legibility and faithfulness, not just technical file validity.

Quality-control checks: file and metadata integrity

Confirm that files open, use the intended format, are named correctly and correspond to the right inventory entries.

Check capture details such as operator, date, equipment, settings, item ID and expected/actual page count.

Use cryptographic checksums to detect later bit-level changes; a matching checksum does not prove a scan was accurate originally.

KEY POINT · Technical integrity, source fidelity and provenance are different things.

Full inspection versus sampling

Critical items or completeness checks may justify 100% inspection; other visual criteria can use risk-based sampling under a documented QA plan.

Sample 10 files out of 100 is not the same as guaranteeing the other 90 are correct; interpret findings in light of selection and error rates.

If error rates exceed project thresholds, pause, investigate and expand inspection or recapture the affected batch.

KEY POINT · Sampling reduces effort only when accompanied by clear risk decisions.

A concrete batch example

A clerk expected 12 pages but received 13 image files: one page may be duplicated or a previously unlisted insert may exist.

The manifest shows page 004 twice and page 011 absent, so the batch cannot pass completeness checks.

Correct the scan and manifest, confirm order, then rerun visual QC and document approval before indexing.

KEY POINT · Page counts alone do not prove completeness; reconcile identities and order.

OCR: useful output, imperfect recognition

Optical character recognition (OCR) turns visible text in images into machine-readable text. It enables searching and can support accessibility.

Performance varies with print quality, script, handwriting, low contrast, layout, language and scanning quality.

An OCR result is a separate derived text layer; do not mistake it for the source image or a verified transcription.

KEY POINT · OCR creates candidates for correction, not automatic proof of accuracy.

OCR verification: a worked example

Source phrase: Records received: 18 July 2024. An OCR draft might read: Rec0rds rece1ved: 18 JuIy 2024.

Typical confusions include 0/O, 1/l/I, punctuation, split columns and missing marginal text.

Correct the text against the source image; document uncertainty rather than inventing a word that cannot be read.

KEY POINT · Verify critical names, dates, numbers and identifiers against the source.

Searchable PDF and transcripts

A searchable PDF may contain the original page image plus an OCR text layer; OCR quality affects search even when the visible image looks correct.

Transcripts and text equivalents can support access, but reading order, tables, headings and language may need manual correction.

For restricted content, a searchable derivative can introduce discovery risks; follow access-control and redaction policy.

KEY POINT · Accessibility and search depend on accurate, structured text, not OCR alone.

Metadata during digitisation

Descriptive: title, subject, creator, date and source collection. Structural: page sequence and relationships.

Technical: format, dimensions, capture settings and software. Administrative: rights, access restrictions and responsible staff.

Preservation-oriented: events, checksums, derivative lineage and actions taken to maintain trustworthy files.

KEY POINT · Metadata gives a file identity, context and a future management pathway.

Preserving provenance and authenticity

Record where a source came from, who authorised digitisation, when it was captured and which processing steps were applied.

For BRIM, a scan of a formal record may need a documented evidential process and retention rules; a scan is not automatically a legal replacement for the original.

For BLISS, a digitised rare-book page needs bibliographic context, collection links, permissions and an accurate edition/leaf identifier.

KEY POINT · Traceability and institutional policy matter more than an attractive thumbnail.

Rights, privacy and sensitive materials

Determine copyright, consent and public-access status before posting digitised material online.

Restricted files require approved storage and user permissions, not just a hidden URL or a no-index instruction.

Redaction should be approved, recorded and verified on a derivative; never erase protected content from the unaltered master without authority.

KEY POINT · Permission to scan, permission to retain and permission to publish are separate questions.

Secure storage and fixity hand-off

Store working captures and protected masters in managed locations with appropriate access control, backup and recovery procedures.

Record a checksum at a documented hand-off and verify it again when moving files or auditing them.

A hash detects a changed byte sequence; it cannot assess missing pages, incorrect attribution or whether an image was badly captured.

KEY POINT · Preservation planning begins when the first master is created.

BLISS case: digitising a local history collection

Goal: make a permission-cleared photograph and leaflet collection discoverable to library users.

Document item identifiers, physical size, condition, content importance, rights, master/access strategy and bibliographic description.

QC checks include missing leaves, cropping, legibility of captions and searchable text from OCR, with original item links.

KEY POINT · The library output is an accessible, contextualised collection, not just image files.

BRIM case: digitising a records series

Goal: provide an approved internal reference copy of a fictional historical administration file, keeping original order and restrictions.

Capture both sides when required, reconcile page sequence, maintain provenance and apply authorised controls to derivatives.

QC checks include completeness, metadata consistency, correct IDs, traceability of re-scans and approval before any distribution.

KEY POINT · The records output must remain controlled, traceable and intelligible.

Practical demonstration: a fictional four-page file

Download the supplied synthetic scan sample and CSV manifest; compare pages by ID, sequence and visual clarity.

Find one simulated capture defect and one OCR transcription error, then record the correction and quality status.

Use the site calculators to estimate capture resolution, expected image data size and a risk-based inspection plan.

KEY POINT · All practice records are invented; no real personal information is used.

Common failures and their remedies

Cropped text: recapture with margins. Glare or blur: stabilise equipment, adjust lighting and focus, then retest.

Wrong page order: reconcile against the inventory. Uncertain OCR: compare with image and mark unresolved readings.

Wrong format or missing metadata: regenerate the derivative or update the manifest without silently changing the capture master.

KEY POINT · Fix the cause, document the rework, and verify the corrected result.

Putting it together: acceptance and hand-off

Before accepting a batch, confirm scope, inventory match, file-opening tests, visual QC, OCR verification, rights and metadata checks.

Keep an exception log with source IDs, defect descriptions, corrective actions, dates, responsible staff and final status.

Package approved masters and derivatives according to institutional storage, retention, access and preservation policy.

KEY POINT · Digitisation is complete only when outputs are checked and responsibly handed over.

Unit 8 summary: reliable digitisation

Plan → prepare → capture → validate → correct → document → preserve → authorise access.

Different material and audiences require different capture decisions; quality is multi-dimensional, and no single scanner setting guarantees success.

Verified files need accurate metadata, organisation and retrieval so their content can be found and understood.

KEY POINT · Good digitisation means a usable, faithful and traceable surrogate, not merely a digital picture.